

AD 502 – Machine Learning

Pre-Reading Material – Lecture 8

Data Preprocessing

Estimated Reading Time: 25–30 minutes

Why This Topic Matters

Data collected from real-world sources is often incomplete, inconsistent, noisy, or unstructured. **Data preprocessing** cleans, transforms, and organizes raw data into a suitable format for analysis and machine learning.

Learning Outcomes

After completing this pre-reading, you should be able to:

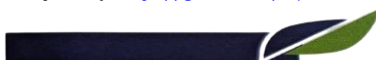
- Explain the importance of **data preprocessing**.
- Identify common problems in raw datasets.
- Understand data cleaning and missing-value handling.
- Explain normalization, standardization, and categorical encoding.
- Understand data reduction and feature engineering.
- Describe the basic preprocessing workflow.

Activate Your Prior Knowledge

1. Missing Data

Suppose a dataset contains an age value recorded as NULL.

Think: Can a machine learning algorithm directly use this missing value?



2. Incorrect Data

Suppose a dataset contains an age of **300 years**.

Think: Should this value be used directly for analysis?

3. Different Formats

Suppose the same country appears as USA and U.S.A.

Think: Why could this inconsistency create problems during analysis?

Prerequisite Concepts

You should be familiar with:

- Python basics
- NumPy
- Pandas DataFrames
- Basic statistics
- Machine learning datasets

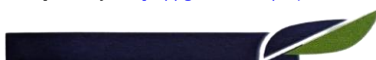
Core Concepts

1. Data Pre-processing

Data preprocessing is the process of **cleaning, transforming, and organizing raw data** into a structured and meaningful format before analysis, statistical modeling, or machine learning.

2. Importance of Data Pre-processing

Preprocessing improves data quality, handles missing information, ensures correct formats, and can improve the performance of machine learning models.



3. Common Problems in Raw Data

Real-world datasets may contain **missing values, duplicate records, noisy data, inconsistent representations, and incorrect data types**. These problems must be addressed before analysis.

4. Major Steps

The five major preprocessing steps are **data cleaning, data integration, data transformation, data reduction, and data discretization**. Each step improves the usability and quality of the dataset.

5. Data Cleaning

Data cleaning involves detecting and correcting errors or inconsistencies. It includes handling missing values, removing duplicate records, correcting inconsistent values, and detecting outliers.

6. Handling Missing Values

Missing values can be handled by removing records or using methods such as **mean, median, mode, KNN, or regression-based imputation**. The appropriate method depends on the data and application.

7. Normalization

Normalization scales numerical values to a specified range. **Min-Max normalization** transforms values using their minimum and maximum values and is useful for algorithms requiring scaled inputs.

8. Standardization

Standardization transforms numerical data so that it has **mean 0 and standard deviation**

1. It is important for algorithms that are sensitive to differences in feature scales.

9. Encoding Categorical Data

Machine learning algorithms often require numerical input, so categorical values need to be converted into numerical representations. For example, categories such as Red, Blue, and Green can be represented numerically.

10. Data Reduction

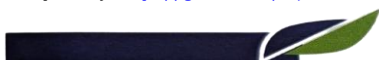
Data reduction decreases dataset size while preserving important information. Techniques include **feature selection, dimensionality reduction such as PCA, and sampling.**

11. Feature Engineering

Feature engineering creates new useful variables from existing data. For example, **Age and Age Group** can be created from a Date of Birth variable.

12. Pre-processing Workflow

A typical workflow is **data collection → data cleaning → data transformation → feature engineering → data reduction → analysis-ready dataset.** Common tools include Pandas, NumPy, Scikit-learn, SQL, R, Excel, and Apache Spark.



13. Practical Example

Consider a dataset containing NULL age values, inconsistent country names, and duplicate records. Preprocessing can replace the missing age, standardize the country name, and remove the duplicate record to produce a clean dataset.

Guided Reading Questions

1. Why is data preprocessing necessary before machine learning?
2. What are the common problems found in raw data?
3. What is data cleaning?
4. How can missing values be handled?
5. What is the difference between normalization and standardization?
6. Why is categorical data encoded?
7. What is the purpose of data reduction?
8. What is feature engineering?
9. What are the major steps in the preprocessing workflow?
10. Why can poor preprocessing lead to unreliable results?

Reflection Activity

Write 100–150 words explaining why a machine learning model trained on **uncleaned data** may produce unreliable results. Consider missing values, duplicate records, inconsistent formats, incorrect values, and feature scaling.

Self-Assessment

1. Data preprocessing is performed before data _____ or machine learning.
2. NULL values represent _____ data.
3. Repeated records in a dataset are called _____ data.
4. Replacing missing numerical values with the average is called _____ imputation.
5. Min-Max is a technique used for _____.

6. Standardization transforms data to have mean _____ and standard deviation _____.
7. Categorical variables are converted into _____ format for machine learning.
8. PCA is an example of _____ reduction.
9. Creating new variables from existing data is called _____ engineering.
10. Removing duplicate records is part of data _____.

Key Takeaway

Data preprocessing is the foundation of reliable machine learning. Cleaning errors, handling missing values, transforming data, reducing unnecessary information, and engineering useful features ensure that the final dataset is accurate, consistent, and suitable for analysis and modeling.

